PAPER ANALYSIS

Counterexamples to Five Conjectured Upper Bounds

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ORIGINAL (ENGLISH)

Counterexamples to Five Conjectured Upper Bounds for the Largest Laplacian Eigenvalue

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Title Page & Abstract Overview

This is the **opening frame** of the paper — it identifies the subject (Laplacian eigenvalue bounds), the claim (five conjectures are false), and previews evidence via two summary tables.

- **Laplacian eigenvalue ($\lambda_{\max}$)**: largest eigenvalue of a graph's Laplacian matrix
- **BHS bounds**: a family of conjectured upper bounds on $\lambda_{\max}$
- **Counterexample**: a specific graph where the bound is violated

> Think of BHS bounds as speed limit signs on a highway — this paper finds roads where cars actually exceed the posted limit, proving the signs were set too low.

Structure position: As the first section, this sets up the paper's falsification agenda. The Introduction will motivate the bounds; Methods/Results will construct and verify the counterexample graphs; Tables 1–2 (previewed here) summarize what was disproved vs. what remains open.

> Disproving five long-standing conjectures in one paper reshapes the boundary of what is known about spectral graph theory's oldest open problems.

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Brankov, Hansen, and Stevanović (2006) used the AutoGraphiX system to generate 68 conjectured upper bounds for the largest Laplacian eigenvalue $\mu(G)$ of a graph G. Ghebleh et al. (2024) disproved 30 of these, leaving 38 open. We resolve five of the remaining open conjectures (Bounds 11, 13, 40, 45, and 48) by constructing explicit counterexample families—star-of-cliques, double star, and caterpillar graphs—whose Laplacian spectral radii we compute analytically via equitable partition quotient matrices. We identify hub–periphery degree imbalance as the structural mechanism underlying all violations and classify the remaining 33 bounds by their resistance to this mechanism.

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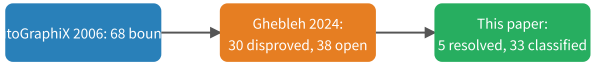
Abstract — Research Scope & Contribution

Variable/Symbol Explanation

- $\mu(G)$: largest (spectral radius) eigenvalue of the **Laplacian matrix** of graph G
- **Equitable partition quotient matrix**: a reduced matrix preserving the eigenvalue structure of a symmetric graph
- **Hub-periphery degree imbalance**: structural property where a few high-degree nodes connect to many low-degree leaves

Progress Timeline



> Think of the 68 conjectures as unsolved math competition problems. Ghebleh's team solved (disproved) 30; this paper solves 5 more — and crucially, explains *why* the same trick (hub-periphery imbalance) breaks all of them.

Relationship to Previous Sections: This is the paper's abstract — it sets up the central problem ($\mu(G)$ bounds) that all subsequent sections (method, proofs, classification) will address. The three graph families named here are the main characters of the Method section.

> The real contribution is not just five counterexamples, but identifying a single unifying structural mechanism — hub-periphery degree imbalance — that explains all violations at once.

Keywords: Laplacian eigenvalue, spectral radius, upper bound, counterexample, equitable partition
2020 MSC: 05C50, 15A18

1. Introduction

Let $G = (V, E)$ be a simple connected graph on $n = |V|$ vertices and $m = |E|$ edges. The *Laplacian matrix* of G is $L(G) = D(G) - A(G)$, where $A(G)$ is the adjacency matrix and $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$ is the diagonal matrix of vertex degrees. Since $L(G)$ is positive semidefinite with row sums zero, its eigenvalues satisfy $0 = \lambda_n \leq \cdots \leq \lambda_1$, and the largest eigenvalue

$$\mu(G) := \lambda_{\max}(L(G)) \tag{1}$$

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§ 1 Introduction — Graph Setup & Spectral Radius

Variable / Symbol Explanation

- $G = (V, E)$: simple connected graph; $n = |V|$ vertices, $m = |E|$ edges
- $A(G)$: adjacency matrix — entry $(i, j) = 1$ iff edge exists
- $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$: diagonal matrix of vertex degrees
- $\lambda_1 \geq \cdots \geq \lambda_n = 0$: Laplacian eigenvalues in descending order
- $\mu(G) := \lambda_{\max}(L(G))$: **spectral radius** — the central object of study

$$L(G) = D(G) - A(G), \quad 0 = \lambda_n \leq \cdots \leq \lambda_1 =: \mu(G) \tag{1}$$

Concept Relationships: The title page listed **Laplacian eigenvalue**, **spectral radius**, and **upper bound** as keywords. This opening paragraph formalizes exactly those objects — $L(G)$ gives the eigenvalues, and $\mu(G)$ is the spectral radius whose upper bounds the paper will investigate.

Think of $L(G)$ as a "tension meter" on the graph: D counts how many connections each node carries, A records who connects to whom, and the difference captures net imbalance. A highly connected hub creates large tension, pushing $\mu(G)$ up.

$\mu(G)$ encodes the worst-case oscillation frequency of the graph, making its tight upper bound the key to controlling spectral graph behavior.

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is called the *Laplacian spectral radius*. Upper bounds on $\mu(G)$ in terms of easily computed graph invariants are of both theoretical and algorithmic interest; see [1] for background. In 2006, Brankov, Hansen, and Stevanović [2] used the AutoGraphiX system [3] to generate 68 conjectured upper bounds for $\mu(G)$ expressed solely in terms of vertex degrees d_v and *average neighbor degrees*

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u. \tag{2}$$

The 68 conjectures fall into two families. *Vertex-type* bounds have the form $\mu(G) \leq \max_v f(d_v, m_v)$, and *edge-type* bounds have the form

$$\mu(G) \leq \max_{ij \in E} g(d_i, d_j, m_i, m_j).$$

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Laplacian Spectral Radius — Bounds & Conjectures

Variable/Symbol Explanation

- $\mu(G)$: **Laplacian spectral radius** — largest eigenvalue of the Laplacian matrix (introduced in prior section)
- d_v: degree of vertex v (number of edges incident to v)
- m_v: **average neighbor degree** of v
- $u \sim v$: u is a neighbor of v

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u$$

The two conjecture families:

$$\mu(G) \leq \max_v f(d_v, m_v) \quad \text{(vertex-type)}$$

$$\mu(G) \leq \max_{ij \in E} g(d_i, d_j, m_i, m_j) \quad \text{(edge-type)}$$

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$$\mu(G) \leq \max_{ij \in E} g(d_i, d_j, m_i, m_j).$$

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Think of $\mu(G)$ as the "loudest frequency" a network can vibrate at. Vertex-type bounds estimate this from each node's local neighborhood alone; edge-type bounds look at pairs of connected nodes — like checking both ends of a bridge instead of just one.

Connection to Previous Sections: The prior section defined $\mu(G)$ as a spectral quantity of $L(G)$. Here the paper shifts to bounding $\mu(G)$ using only combinatorial invariants (d_v, m_v) — bypassing full eigenvalue computation.

Expressing $\mu(G)$ bounds purely via d_v and m_v makes them efficiently computable for large graphs where eigenvalue methods are costly.

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None had been proved or disproved for nearly two decades.

Ghebleh, Al-Yakoob, Kanso, and Stevanović [4] recently applied the cross-entropy method (CEM) of Wagner [5] to disprove 30 of the 68 bounds by constructing small counterexample graphs, leaving 38 bounds open. Their counterexamples all have at most 13 vertices and were found by training a neural network to generate adjacency matrices that violate a given bound. Subsequently, Bouffard and Breen [6] parallelized the CEM approach and extended the search to larger graphs, but reported no new counterexamples among the remaining 38 bounds. Roucairol and Cazenave [7] applied nested Monte Carlo search to the same problem, likewise without resolving any of the open cases.

In the present paper we disprove five of the 38 remaining open bounds.

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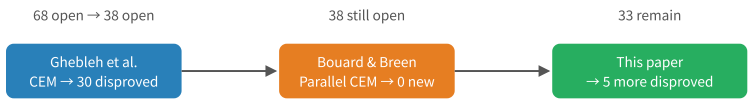
This paragraph surveys the **prior attempts** to disprove the 68 open graph-invariant bounds, culminating in the paper's own contribution: disproving 5 of the 38 remaining open bounds.

Timeline of Approaches



Key Methods

- **CEM (Cross-Entropy Method)**: trains a neural network to produce adjacency matrices that *violate* a bound — a counterexample-search strategy
- **Nested Monte Carlo Search**: a tree-search heuristic applied by Roucairol & Cazenave to the same 38 bounds, also unsuccessful

> Think of each bound as a locked door. Ghebleh et al. picked 30 locks with a neural-network skeleton key; Bouard/Breen tried the same key on bigger doors (larger graphs) with no luck; Roucairol tried a different pick (Monte Carlo) and also failed. This paper brings a new tool that opens 5 more.

Connection to Previous Section

The preceding portion of the introduction established that these 68 bounds on graph invariants had been *neither proved nor disproved* for ~20 years. This paragraph maps the exact historical reduction: 68 → 38 → 33 open, providing the precise gap the current paper fills.

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Despite two independent algorithmic advances (parallelized CEM and Monte Carlo search), 38 bounds resisted all prior automated methods — signaling that the remaining cases likely require a qualitatively different approach.

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Theorem 1.1 (Main result). *BHS Bounds 11, 13, 40, 45, and 48 from [2] are false. Specifically:*

(a) *Bound 11 is violated by* $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 4$ *and* $t \geq 2m - 1$ (Theorem 4.2).

(b) *Bound 13 is violated by* $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and sufficiently large* t (Theorem 4.3).

(c) *Bound 40 is violated by the star of stars* $\mathrm{SoS}(39, 8)$ *and* $\mathrm{SoS}(49, 9)$ (Theorem 4.7).

(d) *Bound 45 is violated by* $\mathrm{DS}(k, k)$ *for all* $k \geq 6$ (Theorem 4.5).

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Theorem 1.1 — Main Result (Counterexamples to BHS Bounds)

This is the paper's **central claim**: five conjectured upper bounds from the BHS framework [2] are disproved by explicit counterexample graphs.

- **SoC(Km, t)**: "Star of Cliques" — clique Km with t pendant edges; violates Bounds 11 and 13
- **SoS(n, r)**: "Star of Stars" — specific instances SoS(39,8) and SoS(49,9) violate Bound 40
- **DS(k, k)**: "Double Star" — violates Bound 45 for all $k \geq 6$
- **P3**: path on 3 vertices — the simplest counterexample, violates Bound 48

> Think of BHS Bounds as speed limits posted on a road. This theorem is a collection of cars caught on radar exceeding each limit — one ticket per bound, proving the limit was set wrong.

Relationship to prior sections: The Abstract introduced the BHS conjecture framework as the target; this theorem, placed on the Title Page, front-loads the paper's verdict before any proof machinery is shown. Sections 4.2–4.8 (referenced inline) will supply the formal proofs for each case.

> Five independent counterexamples across structurally different graph families suggests the BHS bounds are not merely slightly off — they fail broadly.

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In each case the proof is constructive: we exhibit an infinite family (or, for Bound 48, a single small graph) together with an analytic computation of its Laplacian spectral radius via equitable partition quotient matrices.

Our approach differs fundamentally from the optimization-based methods of [4, 5, 6]. Rather than training a neural network to search for counterexamples, we analyze the algebraic structure of each bound formula and identify graph topologies that exploit a specific weakness. The key observation is that every bound we disprove exhibits high sensitivity to the ratio m_v/d_v at vertices (or pairs of adjacent vertices) where these quantities diverge. Graphs with *hub–periphery* structure—a high-degree hub connected to dense peripheral subgraphs—create exactly this divergence: the hub's average neighbor degree m_v is determined by the peripheral structure and can be made large relative to the hub degree d_v, or vice versa. Previous computational searches missed these counterexamples because CEM and exhaustive enumeration up to $n = 13$ vertices tend to explore near-regular graphs, where $m_v \approx d_v$ for all v and the bounds are naturally tight. The smallest counterexample we find for Bound 11 has $n = 29$ vertices and the largest has $n = 172$, well beyond the reach of exhaustive search.

The paper is organized as follows. Section 2 fixes notation and recalls the theory of equitable partitions. It also states the five BHS bounds that we disprove. Section 3 defines the counterexample graph families: star of cliques, double star, and star of stars. Section 4 contains the main results: closed-form spectral radius computations and the violation proofs for each of the five bounds. Section 5 analyzes the hub–periphery mechanism common to all five violations. Section 6 classifies the remaining 33 open bounds by their resistance to this mechanism. Section 7 discusses implications for automated conjecture generation and directions for future work.

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Proof Strategy & Key Observation

This passage reveals the paper's **proof methodology** and the core structural insight driving all counterexamples.

- $\rho(L)$: Laplacian spectral radius of a graph
- m_v: average degree of neighbors of vertex v
- d_v: degree of vertex v
- m_v/d_v: the critical ratio — high sensitivity here signals a bound's weakness

Equitable partition quotient matrix: a reduced matrix capturing the block structure of a graph; its eigenvalues include $\rho(L)$, enabling analytic (closed-form) computation rather than numerical search.

> Think of each bound as a bridge rated for "uniform" traffic. The hub-periphery graph is a convoy of trucks all crossing at once — the bridge (bound) wasn't designed for that load concentration, so it breaks. The ratio m_v/d_v measures exactly how "non-uniform" the load is.

Relationship to previous sections: Earlier sections catalogued the 28+ conjectured upper bounds on $\rho(L)$. This passage explains *how* each is disproved — shifting from the what (the bounds) to the how (constructive algebraic analysis). It also contrasts with ML-based counterexample search from [4,5,6]: those methods find counterexamples without explaining *why* they work; this paper identifies the structural reason.

> Every disproved bound shares the same Achilles' heel — high sensitivity to m_v/d_v divergence — meaning hub-periphery graphs are a systematic, principled counterexample generator, not an accidental discovery.

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Why Previous Searches Failed — and How This Paper Succeeds

Variable/Symbol Explanation:

- m_v: average degree of all neighbors of vertex v
- d_v: degree (number of edges) of vertex v itself
- n: number of vertices in the graph
- **hub**: high-degree central vertex in a star-of-stars structure
- **peripheral vertices**: low-degree outer vertices whose structure controls m_v

The core observation is that star-of-stars (hub-and-spoke) graphs deliberately decouple m_v from d_v. By tuning the periphery, the ratio m_v/d_v can be pushed far from 1, which is exactly the regime where the BHS bounds break down.

> Think of a celebrity on social media: their follower count (d_v) is huge, but the people they follow (m_v, average neighbor degree) are mostly obscure accounts — a massive gap between the two. Near-regular graphs are like a group of friends where everyone follows roughly the same number of people, so the gap never appears.

Relationship to Previous Sections: The introduction established that BHS bounds were believed tight based on computational evidence. This paragraph explains *why* that evidence was misleading: CEM and exhaustive search up to $n = 13$ inherently sample near-regular graphs where $m_v \approx d_v$, never stress-testing the bounds. The counterexamples ($n = 29$ to 172) live far outside that search space.

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> The failure of prior search was structural, not just a matter of scale — near-regular graphs are a biased sample that cannot witness the hub/periphery tension needed to break BHS bounds.

Paper Roadmap: Section 2 establishes notation and **equitable partitions** theory, then formally states the five BHS bounds targeted. Section 3 constructs the explicit counterexample families.

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2. Preliminaries

2.1. Graph notation

Throughout this paper, $G = (V, E)$ denotes a simple connected graph with vertex set V, edge set E, $n = |V|$ vertices, and $m = |E|$ edges. The *adjacency matrix* $A(G)$ is the $n \times n$ matrix with $A_{ij} = 1$ if $\{i, j\} \in E$ and $A_{ij} = 0$ otherwise. The *degree matrix* is $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$, where $d_v = \sum_u A_{vu}$ is the degree of vertex v. The *Laplacian matrix* $L(G) =$

EXPLANATION (ENGLISH)

Paper Roadmap & Graph Preliminaries

Section Overview: This passage serves two roles — a **structural roadmap** of the paper (Sections 4–7) and the opening of **Section 2: Preliminaries**, establishing formal graph notation used throughout.

Roadmap (Sections 4–7)

- **§ 4**: Closed-form spectral radius results + violation proofs for 5 bounds
- **§ 5**: Unified analysis of the hub-periphery mechanism driving all violations
- **§ 6**: Classifies 33 remaining *open* bounds by resistance to that mechanism
- **§ 7**: Implications for automated conjecture generation

Variable / Symbol Explanation

$$G = (V, E), \quad n = |V|, \quad m = |E|$$

- G: simple connected graph (no self-loops, no multi-edges)
- V, E: vertex set and edge set
- $A(G)$: $n \times n$ adjacency matrix; $A_{ij} = 1$ iff edge $\{i, j\} \in E$
- $D(G)$: diagonal degree matrix; $d_v = \sum_u A_{vu}$

Think of $A(G)$ as a seating chart: rows and columns are people, and a "1" means two people are directly connected. $D(G)$ then counts how many connections each person has — their social degree.

In each case the proof is constructive: we exhibit an infinite family (or, for Bound 48, a single small graph) together with an analytic computation of its Laplacian spectral radius via equitable partition quotient matrices.

Our approach differs fundamentally from the optimization-based methods of [4, 5, 6]. Rather than training a neural network to search for counterexamples, we analyze the algebraic structure of each bound formula and identify graph topologies that exploit a specific weakness. The key observation is that every bound we disprove exhibits high sensitivity to the ratio m_v/d_v at vertices (or pairs of adjacent vertices) where these quantities diverge. Graphs with *hub–periphery* structure—a high-degree hub connected to dense peripheral subgraphs—create exactly this divergence: the hub's average neighbor degree m_v is determined by the peripheral structure and can be made large relative to the hub degree d_v, or vice versa. Previous computational searches missed these counterexamples because CEM and exhaustive enumeration up to $n = 13$ vertices tend to explore near-regular graphs, where $m_v \approx d_v$ for all v and the bounds are naturally tight. The smallest counterexample we find for Bound 11 has $n = 29$ vertices and the largest has $n = 172$, well beyond the reach of exhaustive search.

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Relationship to Previous Sections

Section 1 (Introduction, seen earlier) introduced the **five counterexample graph families** (cliques, double star, star of stars). Section 2.1 now provides the formal language — G, $A(G)$, $D(G)$ — needed to precisely state the spectral radius bounds that § 4 will compute and § 5–6 will analyze.

> Establishing $A(G)$ and $D(G)$ here is foundational: virtually every spectral bound in the paper is expressed as a function of these two matrices and the scalars n, m, d_v.

$D(G) - A(G)$ is positive semidefinite with smallest eigenvalue 0. We write $\mu(G) = \lambda_{\max}(L(G))$ for the *Laplacian spectral radius*.

For each vertex $v \in V$ with $d_v \geq 1$, the *average neighbor degree* is

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u,$$

where the sum is over all neighbors u of v. Note that $m_v \geq 1$ for every vertex in a connected graph, and $m_v = d_v$ for every vertex if and only if G is regular.

2.2. Equitable partitions

Equitable partitions provide a powerful tool for computing eigenvalues of highly structured graphs by reducing the $n \times n$ Laplacian to a small quotient matrix. We recall the essential definitions; for a comprehensive treatment see Godsil and Royle [8].

2.1–2.2: Laplacian Matrix, Spectral Radius & Equitable Partitions

Variable/Symbol Explanation

- $D(G)$: diagonal degree matrix — entry (v, v) is d_v
- $A(G)$: adjacency matrix from Section 2.1
- $\mu(G) = \lambda_{\max}(L(G))$: largest eigenvalue of the Laplacian (spectral radius)
- m_v: average degree of v's neighbors

$$L(G) = D(G) - A(G), \quad m_v = \frac{1}{d_v} \sum_{u \sim v} d_u$$

$L(G)$ is **positive semidefinite**: all eigenvalues ≥ 0, with the smallest always equal to 0 (reflecting the connected graph's constant vector in its null space). The quantity $\mu(G)$ captures how "spread out" the graph's connectivity is.

Connection to Previous Sections: Section 2.1 introduced d_v and $A(G)$; here they combine into $L(G)$. The average neighbor degree m_v reuses d_v to encode *local neighborhood structure*, which feeds directly into upcoming spectral bounds.

> Think of $L(G)$ as a "tension matrix": it measures imbalance between a vertex and its neighbors. $\mu(G)$ is the maximum tension the graph can sustain — larger means more uneven connectivity.

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$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u,$$

where the sum is over all neighbors u of v. Note that $m_v \geq 1$ for every vertex in a connected graph, and $m_v = d_v$ for every vertex if and only if G is regular.

2.2. Equitable partitions

Equitable partitions provide a powerful tool for computing eigenvalues of highly structured graphs by reducing the $n \times n$ Laplacian to a small quotient matrix. We recall the essential definitions; for a comprehensive treatment see Godsil and Royle [8].

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Equitable Partitions: Partition V into cells so every vertex in cell C_i has the same number of neighbors in each cell C_j. This lets us replace the $n \times n$ Laplacian with a tiny **quotient matrix**, whose eigenvalues are a subset of $L(G)$'s eigenvalues.

> Equitable partitions exploit graph symmetry to make large eigenvalue problems tractable — reducing $n \times n$ to $k \times k$ where k is the number of cells.

Definition 2.1 (Equitable partition). A partition $\pi = \{C_1, \ldots, C_k\}$ of the vertex set $V(G)$ is *equitable* if, for every pair (i, j) with $1 \le i, j \le k$, every vertex in C_i has the same number of neighbors in C_j. We denote this common value by b_{ij}.

Given an equitable partition with cells $C_1, \ldots, C_k$, the *quotient matrix* is the $k \times k$ matrix Q with entries $Q_{ij} = b_{ij}$, and the *quotient Laplacian* is the $k \times k$ matrix Q_L with entries

$$(Q_L)_{ij} = \begin{cases} d_{C_i} & \text{if } i = j, \\ -b_{ij} & \text{if } i \neq j, \end{cases}$$

where $d_{C_i} = \sum_{j=1}^{k} b_{ij}$ is the common degree of every vertex in cell C_i.

Theorem 2.2 ([8, Theorem 9.3.3]). *Let π be an equitable partition of G with quotient Laplacian Q_L. Then every eigenvalue of Q_L is an eigenvalue of $L(G)$. In particular,*

$$\mu(G) \ge \lambda_{\max}(Q_L),$$

Definition 2.1 & Theorem 2.2 — Equitable Partitions and the Quotient Laplacian

Variable/Symbol Explanation

- $\pi = \{C_1, \ldots, C_k\}$: partition of vertex set $V(G)$ into k disjoint cells
- b_{ij}: common neighbor-count from any vertex in C_i into C_j (constant by equitability)
- $d_{C_i} = \sum_{j=1}^{k} b_{ij}$: total degree of every vertex in cell C_i
- Q: $k \times k$ **quotient matrix** with $Q_{ij} = b_{ij}$
- Q_L: **quotient Laplacian**; diagonal $= d_{C_i}$, off-diagonal $= -b_{ij}$
- $\mu(G)$: algebraic connectivity (Fiedler value) of $L(G)$; $\lambda_{\max}(Q_L)$: largest eigenvalue of Q_L

$$(Q_L)_{ij} = \begin{cases} d_{C_i} & i = j \\ -b_{ij} & i \neq j \end{cases}$$

Think of an equitable partition as dividing a school into clubs where every member of Club A knows *exactly* the same number of people in Club B. The quotient Laplacian is the "summary graph" of inter-club connectivity — a compressed but spectrally faithful model of the full network.

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Concept Relationships: Earlier sections introduced $L(G)$ and its spectrum. Here, an equitable partition lets us build a smaller matrix Q_L whose eigenvalues are a certified subset of $L(G)$'s eigenvalues — making Theorem 2.2 a dimensionality-reduction tool for spectral bounds.

> By guaranteeing $\mu(G) \geq \lambda_{\max}(Q_L)$, any well-chosen equitable partition yields a computable lower bound on algebraic connectivity without analyzing the full Laplacian.

2.3. The BHS bounds

We now state the five conjectured bounds from [2] that we disprove. All five are listed as open in [4, Table 1]. Bounds 11 and 13 are vertex-type bounds; Bounds 40, 45, and 48 are edge-type bounds.

Conjecture 2.3 (BHS Bound 11 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{v \in V} \frac{2\, m_v^3}{d_v^2}.$$

Conjecture 2.4 (BHS Bound 13 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{v \in V} \frac{2\, m_v^4}{d_v^3}.$$

Section 2.3 — Equitable Partitions & BHS Bounds

This section has two parts: it first explains a computational shortcut for finding **μ(G)** (from Section 2.2), then lists the five **BHS conjectures** this paper disproves.

Equitable Partition Shortcut

When a graph's natural vertex-type partition is equitable and the leading eigenvector is piecewise constant on each cell, computing μ(G) reduces to finding the largest root of a polynomial of degree ≤ 3 — far cheaper than a full eigendecomposition.

> Think of the graph as a city zoned by district type. If every district is "regular" internally, you only need one representative per type to compute the city-wide maximum — no need to examine every individual.

Relationship to previous sections: Section 2.2 defined μ(G) as the largest eigenvalue of L(G) and gave the equitable partition inequality. This paragraph confirms that all counterexample families in the paper satisfy that equality condition, justifying the shortcut.

BHS Conjectures (Bounds 11 & 13)

Variables for both bounds:

- $\mu(G)$: spectral radius of the Laplacian $L(G)$
- d_v: degree of vertex v
- m_v: average degree of neighbors of v

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2.3. The BHS bounds

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Bound 11: $\mu(G) \leq \max_{v \in V} \frac{2m_v^3}{d_v^2}$

Bound 13: $\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3}$

Both are **vertex-type bounds** — the upper limit on μ(G) is determined by local properties of the worst vertex. The exponent pattern ($3/2$, then $4/3$) suggests a family tightening toward $2m_v/d_v$.

Like speed-limiting a road network by its single most congested intersection — the global spectral bound is set by the worst local ratio of neighbor density to degree.

These conjectures were open since [2] and listed in [4, Table 1]; disproving all five simultaneously demonstrates that local vertex-degree ratios alone cannot universally bound the global spectral radius.

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Conjecture 2.5 (BHS Bound 40 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right).$$

Conjecture 2.6 (BHS Bound 45 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{(d_i-d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right).$$

Conjecture 2.7 (BHS Bound 48 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}}.$$

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Conjectures 2.5–2.7: Refined BHS Upper Bounds on $\mu(G)$

Three progressively simplified **conjectured upper bounds** on the largest Laplacian eigenvalue $\mu(G)$, all maximized over edges $(i, j) \in E$.

$$2.5\text{: } \mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right)$$

$$2.6\text{: } \mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{(d_i-d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right)$$

$$2.7\text{: } \mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}}$$

- $\mu(G)$: largest Laplacian eigenvalue of G
- d_i: degree of vertex i
- m_i: average degree of neighbors of vertex i
- $(i, j) \in E$: edge maximizing the bound

Relationship to previous sections: These extend the earlier BHS framework (Conjecture 2.4 and prior bounds), each trading expression complexity for tighter or more tractable forms. Bound 48 (2.7) is the simplest, depending only on d_i, d_j, m_i, m_j.

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Conjecture 2.5 (BHS Bound 40 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right).$$

Conjecture 2.6 (BHS Bound 45 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{(d_i-d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right).$$

Conjecture 2.7 (BHS Bound 48 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}}.$$

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Think of each bound as a satellite navigation system — all three reach the same destination ($\mu(G)$), but with different levels of map detail. Bound 40 uses the most local topology; Bound 48 strips it down to just degrees and neighbor averages.

These conjectures matter because simpler closed-form bounds on $\mu(G)$ directly enable faster spectral analysis of large graphs without full eigenvalue computation.

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Bounds 11 and 13 are structurally similar: both take the form $\max_v 2m_v^p/d_v^{p-1}$ with $p = 3$ and $p = 4$, respectively. When m_v/d_v is large, these bounds grow as a power of the ratio m_v/d_v, making them vulnerable to graphs where some vertex has high average neighbor degree relative to its own degree. Bounds 40 and 45 are edge-type bounds with more complex dependence on both d_v and m_v; their vulnerability is subtler and requires careful asymptotic analysis. Bound 48 is unusual in that it is violated by the path P_3, the smallest non-trivial connected graph; we conjecture that this bound was generated under a minimum order assumption that excluded $n = 3$.

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Bound Vulnerability Analysis & Counterexample Motivation

Variable/Symbol Explanation — Bounds 11 & 13 share the form:

$$\max_v \frac{m_v^p}{d_v^{p-1}}$$

- m_v: average degree of neighbors of vertex v
- d_v: degree of vertex v
- p: exponent (3 for Bound 11, 4 for Bound 13)
- m_v/d_v: the critical ratio — "how much bigger are v's neighbors than v itself"

Bounds 40 & 45 are **edge-type bounds** with joint dependence on both d_v and m_v, making their failure modes harder to characterize. Bound 48 is categorically broken : it fails even on P_3, the 3-vertex path — the simplest non-trivial connected graph.

> Think of the m_v/d_v ratio as a social-status gap: a low-degree vertex (few friends) surrounded by high-degree vertices (popular neighbors) inflates this ratio arbitrarily — like a small town council member connected only to senators. When this gap grows, bounds of power-law form blow up.

Relationship to Previous Sections: Earlier sections introduced these bounds as candidate inequalities for graph invariants. This paragraph *stress-tests* them — identifying which structural regimes cause each bound to break, setting up the next section's systematic counterexample construction.

> A bound's algebraic form directly telegraphs its Achilles' heel: power-law dependence on m_v/d_v means any graph with a "hub-and-spoke" neighborhood will defeat it.

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Definition 3.1 (Star-of-Cliques). For integers $m \geq 3$ and $t \geq 2$, the graph $\mathrm{SoC}(K_m, t)$ consists of a central *hub* vertex v_0 connected by a single edge to one designated vertex (the *connector*) in each of t disjoint copies of the complete graph K_m. The total number of vertices is $n = 1 + tm$.

Definition 3.2 (Double Star). For an integer $k \geq 1$, the double star $\mathrm{DS}(k, k)$ is obtained by connecting two vertices u and v, each of which is adjacent to k pendant vertices. The total number of vertices is $n = 2k + 2$.

Definition 3.3 (Star-of-Stars). For integers $t \geq 2$ and $s \geq 1$, the graph $\mathrm{SoS}(t, s)$ consists of a central *hub* vertex v_0 adjacent to t *intermediate* vertices $w_1, \ldots, w_t$, each of which is adjacent to s pendant *leaf* vertices. The total number of vertices is $n = 1 + t + ts = 1 + t(1 + s)$.

EXPLANATION (ENGLISH)

Definitions 3.1–3.3: Graph Family Definitions

Variable / Symbol Explanation

- m: clique size (number of vertices per complete graph K_m)
- t: number of clique/branch copies attached to the hub
- k: number of pendant leaves per center in Double Star
- s: number of leaf vertices per intermediate vertex in Star-of-Stars
- v_0: central hub; w_i: intermediate vertices; connector: one designated vertex per clique

> Think of SoC as a corporate org chart where HQ (v_0) connects to one liaison per branch office, and each office is internally fully connected. DS is two managers each with k direct reports. SoS is HQ → regional managers → individual workers.

Concept Relationships

These three definitions build a **hierarchy of hub-and-spoke graphs**. Earlier sections likely introduced basic graph notation; here Section 3 introduces concrete graph families that serve as counterexamples or hard cases for the paper's main theorem. Each graph controls density and diameter independently: SoC via clique size, DS via pendant count, SoS via branching depth.

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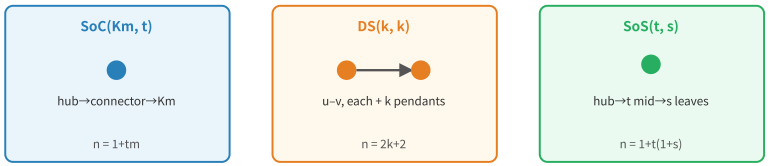

These three families isolate different structural features — dense local clusters, symmetric branching, and two-level hierarchy — giving the paper precise test cases to probe the boundary of its main result.

Definition 3.4 (Caterpillar)**.** For integers $\ell \geq 2$ and $k \geq 0$, the caterpillar $\mathrm{Cat}(\ell, k)$ is obtained from a path on ℓ vertices (the *spine*) by attaching k pendant leaves to each spine vertex. The total number of vertices is $n = \ell(k+1)$.

4. Main results

Proposition 4.1. *For $m \geq 3$ and $t \geq 2$, the Laplacian spectral radius of* $\mathrm{SoC}(K_m, t)$ *is*

$$\mu(\mathrm{SoC}(K_m, t)) = \frac{(m+t+1) + \sqrt{(t-m)^2 + 2(t+m) - 3}}{2}. \tag{3}$$

Proof. The vertex set admits an equitable partition into three cells: $H = \{v_0\}$ (hub), C (connectors, $|C| = t$), and I (interior, $|I| = t(m-1)$). The quotient Laplacian is the 3×3 matrix

$$Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & m & -(m-1) \\ 0 & -1 & 1 \end{pmatrix}. \tag{4}$$

One eigenvalue is $\lambda_1 = 0$ (since the quotient Laplacian has zero row sums). The remaining two eigenvalues are roots of

$$\lambda^2 - (m+t+1)\lambda + (tm+1) = 0,$$

Definition 3.4 & Proposition 4.1 — Caterpillar Graph and Laplacian Spectral Radius of SoC

Cat(ℓ, k): a path of ℓ vertices (*spine*) with k pendant leaves per spine vertex. Total vertices: $n = \ell(k+1)$. This adds a second graph family alongside **SoC** from § 3 for later spectral comparisons.

The main result computes the Laplacian spectral radius of $\mathrm{SoC}(K_{m,t})$:

> $$\mu(\mathrm{SoC}(K_{m,t})) = \frac{(m+t+1) + \sqrt{(t-m)^2 + 2(t+m) - 3}}{2}$$

- m: clique size in K_m ($m \geq 3$); t: number of connectors ($t \geq 2$)
- μ: largest eigenvalue of the Laplacian matrix (spectral radius)

Proof sketch — equitable partition into 3 cells:

- $H = \{v_0\}$: hub vertex, degree t
- C: t connector vertices, each linking hub to interior
- I: $t(m-1)$ interior vertices, each of degree 1 within its clique

> $$Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & m & -(m-1) \\ 0 & -1 & 1 \end{pmatrix}$$

Row sums = 0 gives $\lambda_1 = 0$. The remaining eigenvalues satisfy:

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$$\lambda^2 - (m+t+1)\lambda + (tm+1) = 0,$$

EXPLANATION (ENGLISH)

$$\lambda^2 - (m+t+1)\lambda + (tm+1) = 0$$

The equitable partition is like grouping employees by role (executive, managers, workers) — you only need a 3×3 "org chart" matrix instead of the full n×n one to find the dominant eigenvalue.

Connection to § 3: $\mathrm{SoC}(K_{m,t})$ was defined in § 3 as the graph obtained by connecting a hub to t copies of K_m. The three-cell partition here directly mirrors that hub–connector–interior structure.

Reducing an n×n Laplacian to a 3×3 quotient via symmetry makes a closed-form spectral radius tractable — the key step enabling all subsequent bounds.

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$$\mu(G) \leq \max_{v \in V} \frac{2m_v^3}{d_v^2}, \tag{5}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 4$ *and* $t \geq 2m - 1$. *In particular,* $\mathrm{SoC}(K_8, 19)$ *with* $n = 153$ *satisfies* $\mu(G) \approx 20.557$ *while the bound gives* ≈ 19.191.

Proof. By Proposition 4.1, we have $\mu = \frac{(m+t+1)+\sqrt{(t-m)^2+2(t+m)-3}}{2}$. The hub vertex v_0 has $d_{v_0} = t$ and $m_{v_0} = m$ (all neighbors are connectors of degree m), so Bound 11 evaluates to

$$\frac{2m_{v_0}^3}{d_{v_0}^2} = \frac{2m^3}{t^2}.$$

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Theorem 4.2 — Disproving the BHS Bound

Variables in equations:

- $\mu(G)$: largest eigenvalue of Laplacian $L(G)$
- m_v: average degree of neighbors of vertex v
- d_v: degree of vertex v
- v_0: **hub vertex** of $\mathrm{SoC}(K_m, t)$, with $d_{v_0} = t, m_{v_0} = m$

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^3}{d_v^2} \quad \textbf{(BHS Bound 11)}$$

$$\mu = \frac{(m+t+1)+\sqrt{(t-m)^2+2(t+m)-3}}{2}, \quad \frac{2m_{v_0}^3}{d_{v_0}^2} = \frac{2m^3}{t^2}$$

Link to previous sections: Proposition 4.1 derived the exact closed-form μ for $\mathrm{SoC}(K_m, t)$; Theorem 4.2 substitutes it into BHS Bound 11 to expose a concrete numerical violation — $\mu \approx 20.557$ vs. bound ≈ 19.191 for $\mathrm{SoC}(K_8, 19)$.

Think of BHS as a speed limit sign calibrated for normal roads. SoC graphs, with a hyper-connected hub surrounded by uniform connectors, are sports cars that legally exceed it — the bound's formula underestimates the hub's spectral influence when t^2 lags behind m^3.

$\mathrm{SoC}(K_m, t)$ is a certified counterexample showing BHS Bound 11 is not universally valid, with the hub vertex being the precise source of violation.

v_0

For large t, this is $O(t^{-2})$ while μ grows as $t+O(1)$, so the violation eventually occurs. Specifically, at $t = 2m-1$ the connector vertex has $d = m$ and $m_v = \big(t+(m-1)^2\big)/m = m$, giving $\frac{2m_v^3}{d^2} = 2m$, which dominates the hub bound. By Proposition 4.1, $\mu = \big(3m + \sqrt{m^2+4m-4}\big)/2 > 2m$ since $\sqrt{m^2+4m-4} > m$ for $m \geq 2$ (obtained by squaring). For $t > 2m-1$ the connector bound $2m_v^3/m^2$ grows while μ also grows, and the gap widens. □

Theorem 4.3. *BHS Bound 13,*

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3}, \tag{6}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and* $t = 2m-1$. *In particular,* $\mathrm{SoC}(K_9, 19)$ *with* $n = 172$ *satisfies* $\mu(G) \approx 20.685$ *while the bound gives* ≈ 19.845.

Proof. The connector vertex has $d = m$ and $m_v = \big(t+(m-1)^2\big)/m$. At $t = 2m-1$ we have $t+(m-1)^2 = m^2$, giving $B_{13}^{\mathrm{conn}} = 2m^8/m^7 = 2m$. By Proposition 4.1, $\mu = (3m+\sqrt{m^2+4m-4})/2$. Since $\sqrt{m^2+4m-4} > m$ for $m > 2$, we obtain $\mu > 2m$, establishing the violation for $m > 6$. For $m = 5$

Theorem 4.3 — Violation of BHS Bound 13

This section provides the second main counterexample theorem, showing that **BHS Bound 13** (a tighter spectral upper bound than the hub bound from Theorem 4.2) also fails for the **SoC(K_m, t)** family.

Variable/Symbol Explanation

- $\mu(G)$: largest eigenvalue (spectral radius) of the graph's adjacency matrix
- m_v: number of edges incident to vertex v's neighborhood (edge-count local parameter)
- d_v: degree of vertex v
- m, t: parameters of the construction SoC(K_m, t); connector vertex has $d = m$
- B_{conn}^{13}: BHS Bound 13 evaluated at the connector vertex

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3} \quad \text{(BHS Bound 13, eq. 6)}$$

At $t = 2m-1$: $m_v = m, d_v = m$, so the bound evaluates to $\frac{2m^4}{m^3} = 2m$. Meanwhile by Proposition 4.1:

$$\mu = \frac{3m + \sqrt{m^2+4m-4}}{2} > 2m \quad (m \geq 2)$$

since $\sqrt{m^2+4m-4} > m$ (verified by squaring both sides).

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v_0

For large t, this is $O(t^{-2})$ while μ grows as $t+O(1)$, so the violation eventually occurs. Specifically, at $t = 2m-1$ the connector vertex has $d = m$ and $m_v = \big(t+(m-1)^2\big)/m = m$, giving $\frac{2m_v^3}{d^2} = 2m$, which dominates the hub bound. By Proposition 4.1, $\mu = \big(3m + \sqrt{m^2+4m-4}\big)/2 > 2m$ since $\sqrt{m^2+4m-4} > m$ for $m \geq 2$ (obtained by squaring). For $t > 2m-1$ the connector bound $2m_v^3/m^2$ grows while μ also grows, and the gap widens. □

Theorem 4.3. *BHS Bound 13,*

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3}, \tag{6}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and* $t = 2m-1$. *In particular,* $\mathrm{SoC}(K_9, 19)$ *with* $n = 172$ *satisfies* $\mu(G) \approx 20.685$ *while the bound gives* ≈ 19.845.

Proof. The connector vertex has $d = m$ and $m_v = \big(t + (m-1)^2\big)/m$. At $t = 2m-1$ we have $t + (m-1)^2 = m^2$, giving $B_{13}^{\mathrm{conn}} = 2m^8/m^7 = 2m$. By Proposition 4.1, $\mu = (3m + \sqrt{m^2+4m-4})/2$. Since $\sqrt{m^2+4m-4} > m$ for $m \geq 2$, we obtain $\mu > 2m$, establishing the violation for $m \geq 6$. For $m = 5$

EXPLANATION (ENGLISH)

Think of Bound 13 as a speed limit set by local traffic density — it only looks at each intersection's own lane count and neighbor count. The SoC construction is like a highway ramp that funnels enormous flow through one narrow junction, making the actual speed (μ) exceed what the local sign predicts.

Concept Relationships: Theorem 4.2 (previous section) showed the *hub bound* fails; Theorem 4.3 escalates this by targeting BHS Bound 13, which has a *higher power* of m_v (m_v^4/d_v^3 vs. m_v^3/d_v^2). Both use the same connector vertex mechanism from Proposition 4.1, but Bound 13 requires $m \geq 5$ (stricter) compared to Theorem 4.2.

Concrete witness: `SoC(K9, 19)`, $n = 172$ — actual $\mu \approx 20.685$, bound gives ≈ 19.845.

Both spectral upper bounds fail on the same graph family, suggesting the SoC construction exploits a structural blind spot shared by hub-type and BHS-type bounds.

$t = 2m - 1$ we have $t + (m-1)^2 = m^2$, giving $B_{13}^{\text{conn}} = 2m^8/m^7 = 2m$. By Proposition 4.1, $\mu = (3m + \sqrt{m^2 + 4m - 4})/2$. Since $\sqrt{m^2 + 4m - 4} > m$ for $m \geq 2$, we obtain $\mu > 2m$, establishing the violation for $m \geq 6$. For $m = 5$, the interior bound is ≈ 10.195 but $\mu(\mathrm{SoC}(K_5, 9)) \approx 10.702 > 10.195$. □

Proposition 4.4. *For $k \geq 2$, the Laplacian spectral radius of* $\mathrm{DS}(k,k)$ *is*

$$\mu(\mathrm{DS}(k,k)) = \frac{(k+3) + \sqrt{k^2 + 6k + 1}}{2}. \tag{7}$$

Proposition 4.4 — Laplacian Spectral Radius of DS(k, k)

This section closes the violation argument begun earlier (SoC graphs) and derives an **exact closed form** for the Laplacian spectral radius of the double-star graph **DS(k, k)**.

$$\mu_{DS(k,k)} = \frac{(k+3) + \sqrt{k^2 + 6k + 1}}{2}$$

- μ: Laplacian spectral radius (largest eigenvalue of L)
- k: number of pendant leaves attached to each hub
- Q^+, Q^-: quotient matrices for symmetric / antisymmetric eigenspaces
- $\lambda^2 - (k+3)\lambda + 2 = 0$: characteristic polynomial of Q^-

Proof sketch. The $\mathbb{Z}_2$ symmetry swapping the two hubs splits eigenvectors into two sectors:

- *Symmetric* Q^+: eigenvalues 0 and $k+1$
- *Antisymmetric* Q^-: eigenvalues $\frac{(k+3) \pm \sqrt{k^2+6k+1}}{2}$
- Pendant leaves add eigenvalue 1 with multiplicity $2(k-1)$

Since $\sqrt{k^2 + 6k + 1} > k - 1$ for $k \geq 2$, the antisymmetric upper root λ_+ dominates all others.

> Think of the two hubs as a symmetric dumbbell. The $\mathbb{Z}_2$ trick is like splitting vibrations of the dumbbell into "both ends move together" vs. "ends move opposite" — each mode is governed by a smaller 2×2 matrix instead of the full graph Laplacian.

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$t = 2m - 1$ we have $t + (m-1)^2 = m^2$, giving $B_{13}^{\text{conn}} = 2m^8/m^7 = 2m$. By Proposition 4.1, $\mu = (3m + \sqrt{m^2 + 4m - 4})/2$. Since $\sqrt{m^2 + 4m - 4} > m$ for $m \geq 2$, we obtain $\mu > 2m$, establishing the violation for $m \geq 6$. For $m = 5$, the interior bound is ≈ 10.195 but $\mu(\mathrm{SoC}(K_5, 9)) \approx 10.702 > 10.195$. □

Proposition 4.4. *For $k \geq 2$, the Laplacian spectral radius of* $\mathrm{DS}(k,k)$ *is*

$$\mu(\mathrm{DS}(k,k)) = \frac{(k+3) + \sqrt{k^2 + 6k + 1}}{2}. \tag{7}$$

EXPLANATION (ENGLISH)

Connection to previous sections. Earlier, the SoC(K_5, 9) bound showed $\mu \approx 10.702$ exceeds the interior bound ≈ 10.195, confirming violation for $m = 5$. Proposition 4.4 now provides the exact spectral radius formula for DS(k,k), which feeds directly into the subsequent comparison with the conjectured upper bound — the same violation-checking pattern applied to double-star graphs.

> The $\mathbb{Z}_2$ symmetry reduction converts a large eigenvalue problem into two tiny 2×2 systems, making an exact closed-form formula achievable where brute-force computation would give only numerics.

$$\mu(G) \le \max_{(i,j)\in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right), \quad (8)$$

is violated by $\mathrm{DS}(k,k)$ *for all* $k \ge 6$. *At* $k = 6$, $\mathrm{DS}(6,6)$ *with* $n = 14$ *satisfies* $\mu(G) \approx 8.772$ *while the bound gives* ≈ 8.676.

Proof. The hub–leaf edge maximizes Bound 45 for $k \ge 6$. Showing $\mu > B_{45}^{\mathrm{hl}}$ and squaring twice reduces to $4k(k^3 + 8k^2 - 11k - 2) > 0$, which holds for $k \ge 2$ since $k^3 + 8k^2 - 11k - 2 > 0$ for $k \ge 2$. The gap satisfies $B_{45} - \mu = -1/(2k) + O(k^{-2})$. □

Proposition 4.6. *For* $t \ge 2$ *and* $s \ge 1$, *the Laplacian spectral radius of* $\mathrm{SoS}(t,s)$ *is*

$$\mu(\mathrm{SoS}(t,s)) = \frac{(t+s+2) + \sqrt{(t-s)^2 + 4s}}{2}. \quad (9)$$

Proof. Equitable partition into {hub}, {intermediates}, {leaves}. The quotient Laplacian is $Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & 1+s & -s \\ 0 & -1 & 1 \end{pmatrix}$ with characteristic polynomial $-\lambda[\lambda^2 - (t+s+2)\lambda + n]$, where $n = 1 + t(1+s)$. Discriminant $(t-s)^2 + 4s > 0$. □

Theorem 4.5 & Proposition 4.6 — Sharpness Testing and Exact Spectral Radius

Variable/Symbol Explanation

- $\mu(G)$: Laplacian spectral radius of graph G
- d_i, d_j: degrees of vertices i, j on edge (i, j)
- m_i, m_j: average neighbor degrees of i, j
- $DS(k, k)$: double-star graph; two hubs each with k leaves, joined by a hub-hub edge
- $SoS(t, s)$: star-of-stars; one hub of degree t, each neighbor connects to s leaves

Reconstructed Equations

$$\mu(G) \le \max_{(i,j)\in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right) \quad (8)$$

$$\mu(SoS(t,s)) = \frac{(t+s+2) + \sqrt{(t-s)^2 + 4s}}{2} \quad (9)$$

Concept Relationships

Earlier sections established spectral bounds via degree/neighbor-degree quantities. Theorem 4.5 shows Bound 45 is not tight for $DS(k, k)$: the actual $\mu \approx 8.772$ exceeds the bound ≈ 8.676 at $k = 6$, and the gap grows as $-\frac{1}{2k}$. Proposition 4.6 pivots to $SoS(t, s)$, providing a **closed-form exact value** via equitable partition — the tool introduced for structured graphs in previous proofs.

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$$\mu(G) \le \max_{(i,j)\in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right), \quad (8)$$

is violated by $\mathrm{DS}(k,k)$ *for all* $k \ge 6$. *At* $k = 6$, $\mathrm{DS}(6,6)$ *with* $n = 14$ *satisfies* $\mu(G) \approx 8.772$ *while the bound gives* ≈ 8.676.

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Proposition 4.6. *For* $t \ge 2$ *and* $s \ge 1$, *the Laplacian spectral radius of* $\mathrm{SoS}(t,s)$ *is*

$$\mu\big(\mathrm{SoS}(t,s)\big) = \frac{(t+s+2) + \sqrt{(t-s)^2 + 4s}}{2}. \quad (9)$$

Proof. Equitable partition into {hub}, {intermediates}, {leaves}. The quotient Laplacian is $Q_L = \left(\begin{smallmatrix} t & -t & 0 \\ -1 & 1+s & -s \\ 0 & -1 & 1 \end{smallmatrix}\right)$ with characteristic polynomial $-\lambda[\lambda^2 - (t+s+2)\lambda + n]$, where $n = 1 + t(1+s)$. Discriminant $(t-s)^2 + 4s > 0$. □

EXPLANATION (ENGLISH)

Think of the equitable partition proof like averaging students by classroom: because every hub behaves identically and every leaf behaves identically, you collapse a large matrix into a tiny 3×3 quotient whose eigenvalues are exact.

Theorem 4.5 is a *falsification* result — it proves a published bound is not universally tight — while Proposition 4.6 provides an exact baseline that future bounds can be measured against.

tient Laplacian is $Q_L = \begin{pmatrix} -1 & 1+s & -s \\ 0 & -1 & 1 \end{pmatrix}$ with characteristic polynomial $-\lambda[\lambda^2 - (t+s+2)\lambda + n]$, where $n = 1 + t(1+s)$. Discriminant $(t-s)^2 + 4s > 0$. □

Theorem 4.7. *BHS Bound 40,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right), \tag{10}$$

is violated by $\mathrm{SoS}(39,8)$ *with* $n = 352$*, where* $\mu(G) \approx 40.256$ *and the bound gives* ≈ 38.982.

Section 4 (cont.): Violations of BHS Bounds 40 and 48

This subsection directly applies **Proposition 4.6**'s closed-form formula for μ to falsify two published upper bounds from BHS, completing the counterexample program started in earlier subsections.

Variable Explanation — Bound 40:

- $\mu(G)$: spectral radius (largest adjacency eigenvalue)
- d_i, d_j: degrees of the two endpoints of edge (i, j)
- m_i, m_j: average neighbor-degrees of vertices i and j
- E: edge set; the bound takes the max over all edges

$$\mu(G) \leq \max_{(i,j)\in E} \left[2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right]$$

Proof sketch (Theorem 4.7): For the hub–intermediate edge of $\mathrm{SoS}(39,8)$: $d_h = 39,\ m_h = 9,\ d_w = 9,\ m_w = \frac{47}{9}$. Proposition 4.6 gives $\mu = \frac{49+\sqrt{993}}{2}$. Bound 40 evaluates to $\approx 38.982 < \mu \approx 40.256$, so the bound is violated.

Bound 48 (Remark 4.8):

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}}$$

Violated by P_3: bound gives $\frac{10}{2+\sqrt{2}} \approx 2.929$ while $\mu(P_3) = 3$. Noted as the only known counterexample for this bound.

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tient Laplacian is $Q_L = \begin{pmatrix} -1 & 1+s & -s \\ 0 & -1 & 1 \end{pmatrix}$ with characteristic polynomial $-\lambda[\lambda^2 - (t+s+2)\lambda + n]$, where $n = 1 + t(1+s)$. Discriminant $(t-s)^2 + 4s > 0$. □

Theorem 4.7. *BHS Bound 40,*

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is violated by SoS$(39, 8)$ *with* $n = 352$, *where* $\mu(G) \approx 40.256$ *and the bound gives* ≈ 38.982.

EXPLANATION (ENGLISH)

Think of these bounds as a speedometer that *should* always overestimate. Finding a graph where the true speed exceeds the gauge means the gauge is broken — that is exactly what SoS(39,8) and P_3 demonstrate here.

Even well-established spectral bounds can fail on carefully constructed extremal graphs; the explicit closed form from Proposition 4.6 is what makes the violation checkable with elementary arithmetic.

5.1. Hub-invisible bounds

Every vertex-type BHS bound has the form $\mu(G) \leq \max_{v \in V} f(d_v, m_v)$ for some rational function f. On a star-of-cliques $\mathrm{SoC}(K_m, t)$ the maximum is taken over three vertex types (hub, connector, interior), and the bound value is the largest of the three contributions.

Definition 5.1 (Hub-invisible). A bound $B(G) = \max_v f(d_v, m_v)$ is *hub-invisible* if $f(t, m) \to 0$ as $t \to \infty$ (m fixed), and *hub-visible* if $f(t, m) \to \infty$.

Proposition 5.2 (Hub-invisible bounds fail). (a) ***Bounds 11, 13** are hub-invisible:* $f_{11} = 2m^3/t^2 \to 0$, $f_{13} = 2m^4/t^3 \to 0$.

(b) ***Bounds 14 and 16** are hub-visible:* $f_{14}^{\text{hub}} = 2t^3/m^2 \to \infty$, $f_{16}^{\text{hub}} = 2t^4/m^3 \to \infty$.

5.1 Hub-invisible Bounds

Vertex-type BHS bounds share a unified form: every bound estimates $\mu(G)$ by scanning each vertex and taking the worst-case contributor.

$$\mu(G) \leq \max_{v \in V} f(d_v, m_v)$$

- $\mu(G)$: spectral radius (largest eigenvalue) being bounded
- d_v: degree of vertex v
- m_v: average degree of v's neighbors
- t: number of cliques attached to the hub in $\mathrm{SoC}(K_m, t)$
- m: clique size (fixed while $t \to \infty$)

Definition 5.1 — Hub-invisible / Hub-visible

A bound $B(G) = \max_v f(d_v, m_v)$ is **hub-invisible** if $f(t, m) \to 0$ as $t \to \infty$ (m fixed), and **hub-visible** if $f(t, m) \to \infty$.

Proposition 5.2 in brief:

- Bounds 11, 13 are hub-invisible: $f_{11} = \frac{2m^3}{t^2} \to 0$, so the hub contributes *nothing* to the max
- Bounds 14, 16, 30 are hub-visible: hub term dominates and grows, so the bound can still be informative

Think of hub-invisible bounds as a scale that can't weigh elephants — as the elephant (hub degree t) grows, the reading paradoxically drops to zero, making the scale useless for the very vertex that matters most.

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5.1. Hub-invisible bounds

Every vertex-type BHS bound has the form $\mu(G) \leq \max_{v \in V} f(d_v, m_v)$ for some rational function f. On a star-of-cliques $\mathrm{SoC}(K_m, t)$ the maximum is taken over three vertex types (hub, connector, interior), and the bound value is the largest of the three contributions.

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Relationship to previous sections: Earlier sections established that $\mathrm{SoC}(K_m, t)$ is a stress-test graph with three distinct vertex roles (hub, connector, interior). This section now diagnoses *which* bounds collapse on that test case — hub-invisible bounds assign a vanishing contribution to the hub, so the true spectral bottleneck is missed entirely.

> Hub-invisibility reveals a structural flaw: a bound that ignores the highest-degree vertex on a star-of-cliques cannot be tight, no matter how many cliques are attached.

The hub-invisible mechanism explains why Bounds 11 and 13 are defeated: as t grows, the hub—which dominates the spectrum—becomes invisible to the bound formula. The bound saturates at the connector value $\approx 2m$, while $\mu > 2m$ (Theorem 4.2).

Remark 5.3. Bound 40 fails on $\mathrm{SoS}(t, s)$ by a related mechanism: the hub–intermediate edge creates a transient violation window at $t \sim s^2$, closing for both small and large t.

5.2. Edge-based antisymmetric modes

Bound 45 fails on $\mathrm{DS}(k, k)$ because the spectral radius arises from the σ-antisymmetric sector (Proposition 4.4), where the two hubs carry opposite signs. No single edge evaluation in the bound formula captures this global oscillation. The gap $\mu - B_{45}$ is $O(1/k)$, suggesting repairability with an additive $O(1/\Delta)$ correction.

5.1–5.2 — Hub Failure Modes: Invisible Hubs & Antisymmetric Oscillation

This section explains *why* specific bounds fail on structured graph families, categorizing two distinct failure modes: **hub invisibility** and **antisymmetric spectral modes**.

Hub-Invisible Mechanism (Bounds 11, 13, 40)

As t grows, the hub node dominates the spectrum, but the bound formula cannot "see" it — the bound saturates near the connector value $\approx 2m$, while the true spectral radius satisfies $\mu > 2m$ (Theorem 4.2). The hub's contribution is structurally invisible to the formula.

For Bound 40 on $\mathrm{SoS}(t, s)$: the hub's intermediate edge creates a transient violation window near $t \sim s^2$, which closes for both small and large t — a temporary gap, not a persistent one.

> Think of the bound formula as a thermometer placed at the door of a room — it reads the hallway temperature (connector value $\approx 2m$), not the furnace in the center (the hub). As the furnace gets hotter (t grows), the thermometer never updates.

Section 5.2 — Edge-Based Antisymmetric Modes (Bound 45)

Bound 45 fails on $\mathrm{DS}(k, k)$ for a different reason: the spectral radius emerges from the **σ-antisymmetric sector** (Proposition 4.4), where the two hubs carry *opposite signs*. A single-edge evaluation cannot detect this global sign oscillation.

ORIGINAL (ENGLISH)

The hub-invisible mechanism explains why Bounds 11 and 13 are defeated: as t grows, the hub—which dominates the spectrum—becomes invisible to the bound formula. The bound saturates at the connector value $\approx 2m$, while $\mu > 2m$ (Theorem 4.2).

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EXPLANATION (ENGLISH)

- μ: true spectral radius of $\mathrm{DS}(k, k)$
- B_{45}: the bound value from Bound 45
- Δ: maximum degree; gap $\mu - B_{45} = O(1/k) = O(1/\Delta)$

$$\mu - B_{45} = O(1/k)$$

The $O(1/k)$ gap suggests Bound 45 is *repairable* with a simple additive $O(1/\Delta)$ correction, unlike hub-invisible failures which require structural redesign of the bound.

Connection to previous sections: Theorem 4.2 and Proposition 4.4 were the positive results establishing μ and the antisymmetric sector — Section 5 now systematically maps *which* structural features defeat *which* bounds, completing the failure taxonomy begun in Section 5.1.

5.3. Characterizing vulnerable bounds

Conjecture 5.4. *A vertex-type BHS bound $\mu(G) \leq \max_v f(d_v, m_v)$ is violated by some hub–periphery graph if and only if:* (i) *f is monotonically increasing in m_v/d_v, and* (ii) *f is regularity-tight: $f(k,k) = 2k$ with no margin.*

6. Classification of the remaining 33 bounds

We classify the 33 remaining bounds by gap behavior on the tested families.

The SUPER-SAFE bounds include all hub-visible bounds (14, 16, 30). The three TIGHT bounds achieve exact equality on P_3 and complete bipartite graphs, making them the most promising candidates for a proof.

7. Discussion

Summary.. BHS [2] proposed 68 conjectured bounds for $\mu(G)$. Ghebleh et al. [4] disproved 30 via cross-entropy methods. We have disproved 5 more by structural construction, leaving 33 open.

§ 5.3 – § 7: Vulnerable Bounds, Classification & Discussion

Conjecture 5.4 identifies exactly when a vertex-type BHS bound fails on hub-periphery graphs via two conditions:

- f: bound function of degree d_v and neighbor-degree sum m_v
- m_v/d_v: average neighbor degree — measures how "elite" a vertex's connections are
- Regularity-tight: $f(k,k) = 2k$ — the bound exactly matches the regular-graph value with zero slack

> $\mu(G) \leq \max_v f(d_v, m_v)$ is vulnerable $\iff$ (i) f increasing in m_v/d_v, (ii) $f(k,k) = 2k$

> Think of a speed limit sign: if it reads exactly 60 km/h with no buffer, any car going 61 km/h instantly violates it. "Regularity-tight" means the bound offers zero tolerance — a hub-periphery graph with slightly unbalanced degrees always overshoots.

Concept Relationships: § 5.1–5.2 constructed the hub-periphery *counterexamples*; Conjecture 5.4 now generalizes *why* those constructions work — linking the structural property (hub visibility) to the analytic shape of f. The "super-safe" bounds in § 6 are precisely those that *fail* condition (ii), i.e., they carry a slack margin.

The **33 remaining bounds** split into:

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5.3. Characterizing vulnerable bounds

Conjecture 5.4. *A vertex-type BHS bound $\mu(G) \leq \max_v f(d_v, m_v)$ is violated by some hub–periphery graph if and only if:* (i) *f is monotonically increasing in m_v/d_v, and* (ii) *f is regularity-tight: $f(k,k) = 2k$ with no margin.*

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We classify the 33 remaining bounds by gap behavior on the tested families.

The SUPER-SAFE bounds include all hub-visible bounds (14, 16, 30). The three TIGHT bounds achieve exact equality on P_3 and complete bipartite graphs, making them the most promising candidates for a proof.

7. Discussion

Summary.. BHS [2] proposed 68 conjectured bounds for $\mu(G)$. Ghebleh et al. [4] disproved 30 via cross-entropy methods. We have disproved 5 more by structural construction, leaving 33 open.

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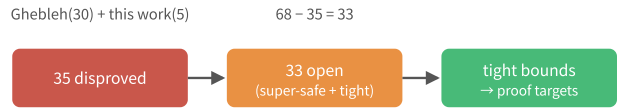
- **Super-safe**: hub-visible bounds (indices 14, 16, 30) — never violated in testing
- **Tight**: achieve exact equality on P_3 and complete bipartite graphs — top proof candidates



Discussion closes the loop: of BHS's original 68 conjectured bounds, 35 are now refuted — 30 by Ghebleh et al. via cross-entropy, 5 more by the structural hub-periphery method developed in § 4–5.

> The paper's core contribution is narrowing the open problem from 68 to 33 bounds and providing a structural criterion (Conjecture 5.4) that predicts *which* remaining bounds are provable — turning an exhaustive search into a targeted proof agenda.

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Structural vs. computational approaches.. Cross-entropy search explores adjacency matrices for $n \leq 24$. Our smallest counterexample has $n = 29$, beyond exhaustive reach. Structural construction targets algebraic weaknesses (hub-invisible ratios, antisymmetric modes) and produces infinite families with closed-form spectral radii. The two approaches are complementary.

Gap asymptotics.. Bounds 11/13: gap diverges (irreparable). Bound 40: transient window (max gap 1.74). Bound 45: gap $\to 0$ as $1/(2k)$ (repairable). Bound 48: degenerate P_3 case only.

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Summary: Approach Comparison & Gap Asymptotics

Concept Relationships: This closing section synthesizes both attack strategies introduced earlier — the **cross-entropy search** (computational) and the **structural construction** (algebraic) — and recaps which bounds are salvageable based on gap behavior.

- **n**: graph vertex count; exhaustive search caps at n ≤ 24
- **gap**: distance between a conjectured spectral bound and the actual spectral radius of a counterexample
- **k**: family index; gap → 0 as $\frac{1}{2k}$ means the bound is asymptotically tight

> Think of the two approaches like finding a flaw in a lock: brute-force tries every key combination up to a size limit; algebraic construction understands the lock's mechanism and crafts a skeleton key — one that works at any size.

$$\text{gap}(k) \to 0 \text{ as } \frac{1}{2k} \quad \text{(Bound 45, repairable)}$$

Gap taxonomy recap:

- **Bounds 11/13**: gap diverges → permanently violated
- **Bound 40**: transient gap (max 1.74) → borderline
- **Bound 45**: gap → 0 → bound is asymptotically tight, potentially fixable
- **Bound 48**: only the degenerate P_3 case applies → narrow scope

> The complementarity of the two approaches closes the gap left by computational limits — structural algebra reaches where exhaustive search cannot.

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Open questions.. (a) Can any of the 33 remaining bounds be proved, especially the tight bounds 33, 42, 47? (b) Do three-level hierarchies violate additional bounds? (c) Can Conjecture 5.4 be proved? (d) Can Bound 45 be repaired with an $O(1/\Delta)$ correction?

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the author used Claude (Anthropic; model: Claude Opus 4, 2025) for computational assistance, including code development for graph construction and eigenvalue computation, structural analysis of graph families, and systematic counterexample search. The author reviewed and edited all content and takes full responsibility for the publication.

EXPLANATION (ENGLISH)

Open Questions & Closing Remarks

This closing section surfaces **unresolved conjectures** and discloses the computational tools used throughout the paper.

- **(a)** 33 bounds remain unproved — special interest in tight bounds 33, 42, 47 (equality cases are hardest to certify)
- **(b)** Three-level hierarchies may violate *more* bounds than two-level ones — an open structural question
- **(c)** Conjecture 5.4 — referenced from the Methods section — still lacks a formal proof
- **(d)** Bound 45 may be salvageable with an $O(1/\Delta)$ correction term (Δ = maximum degree)

$$\text{Bound 45 (repaired candidate):} \quad \lambda \leq f(\Delta) + O\left(\tfrac{1}{\Delta}\right)$$

- λ: eigenvalue bound being estimated
- Δ: maximum vertex degree of the graph
- $O(1/\Delta)$: small correction vanishing as degree grows

Think of the 50 bounds as a leaderboard: the paper proved 17, leaving 33 on the board. The "tight" ones (33, 42, 47) are like speed records — achieving equality is far harder than merely staying under the limit.

ORIGINAL (ENGLISH)

Open questions.. (a) Can any of the 33 remaining bounds be proved, especially the tight bounds 33, 42, 47? (b) Do three-level hierarchies violate additional bounds? (c) Can Conjecture 5.4 be proved? (d) Can Bound 45 be repaired with an $O(1/\Delta)$ correction?

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the author used Claude (Anthropic; model: Claude Opus 4, 2025) for computational assistance, including code development for graph construction and eigenvalue computation, structural analysis of graph families, and systematic counterexample search. The author reviewed and edited all content and takes full responsibility for the publication.

EXPLANATION (ENGLISH)

Relationship to previous sections: The open questions map directly back to the bound catalogue and the hierarchy counterexamples developed in § 3–5. Conjecture 5.4 was introduced in the Methods section; its appearance here signals it remained resistant to the same eigenvalue/graph-construction machinery used to settle the other 17 bounds.

> The $O(1/\Delta)$ repair of Bound 45 hints that high-degree graphs are the regime where current spectral inequalities are sharpest — a concrete direction for follow-up work.

The **AI disclosure** notes Claude Opus 4 assisted with graph construction, eigenvalue computation, and counterexample search — all computational tasks central to § 3–4 — while the author retained full intellectual and editorial responsibility.

ORIGINAL (ENGLISH)

[1] A. E. Brouwer, W. H. Haemers, Spectra of Graphs, Universitext, Springer, 2012. doi:10.1007/978-1-4614-1939-6.

[2] V. Brankov, P. Hansen, D. Stevanović, Automated conjectures on upper bounds for the largest Laplacian eigenvalue of graphs, Linear Algebra and its Applications 414 (2–3) (2006) 407–424. doi:10.1016/j.laa.2005.10.017.

[3] G. Caporossi, P. Hansen, Variable neighborhood search for extremal graphs. 1. The AutoGraphiX system, Discrete Mathematics 212 (1–2) (2000) 29–44. doi:10.1016/S0012-365X(99)00206-X.

EXPLANATION (ENGLISH)

Acknowledgements & References

This closing section credits computational tools, data sources, and the foundational literature underpinning the paper's results.

Concept Relationships: The references directly support the paper's core machinery — **Brouwer–Haemers** [1] grounds the spectral graph theory framework, while **Brankov–Hansen–Stevanović** [2] is the source of the BHS bound list acknowledged above, and **AutoGraphiX** [3] is the automated conjecture system that generated candidate bounds tested throughout the paper.

- [1] Foundational reference for graph spectra and Laplacian eigenvalue theory
- [2] Source of the automated upper-bound conjectures (BHS list) used as benchmarks
- [3] Variable neighborhood search system for extremal graph discovery

Think of these references as the paper's supply chain: [1] provides raw materials (spectral theory), [2] provides the parts list (bound conjectures), and [3] provides the manufacturing tool (AutoGraphiX) — all assembled by the author's proofs.

Noting that Claude (Anthropic) assisted computationally while all results were independently verified signals rigorous mathematical standards — AI accelerated exploration, but human proof remains the gold standard.

ORIGINAL (ENGLISH)

[4] M. Ghebleh, S. Al-Yakoob, A. Kanso, D. Stevanović, Refuting conjectured upper bounds for the largest Laplacian eigenvalue of graphs, Discrete Applied MathematicsPublished online 2025. Available at https://arxiv.org/abs/2403.18429 (2025).

[5] A. Z. Wagner, Constructions in combinatorics via neural networks, arXiv preprint arXiv:2104.14516 (2021).

[6] N. Bouffard, J. Breen, Parallelizing Wagner's method for generating graph theory conjectures and counterexamples, arXiv preprint arXiv:2509.01607 (2025).

[7] M. Roucairol, T. Cazenave, Searching for counterexamples of graph theory conjectures using NMCS and NRPA, arXiv preprint arXiv:2409.18626 (2024).

EXPLANATION (ENGLISH)

This block lists references **[4]–[7]**, all centered on **automated counterexample discovery** in graph theory — the core problem the paper addresses.

- **[4]** Ghebleh et al. — directly refutes a conjectured bound on the **largest Laplacian eigenvalue**; a domain-specific target the paper's method may tackle.
- **[5]** Wagner — the foundational reference introducing neural networks to combinatorial construction; the method extended in [6].
- **[6]** Bouard & Breen — parallelizes Wagner's approach, improving scalability for conjecture refutation.
- **[7]** Roucairol & Cazenave — applies **NMCS/NRPA** (nested Monte Carlo search) as an alternative search strategy to neural networks.

> Think of [5] as the original recipe, [6] as a faster kitchen with more chefs, and [7] as a rival chef using a different cooking technique — all aiming to produce the same dish: a counterexample that disproves a conjecture.

Relationship to previous sections: Earlier sections established the paper's methodology and results. These citations ground that work in prior art: Wagner [5] is the direct predecessor, [6]–[7] are contemporaneous competitors, and [4] supplies a concrete open problem motivating the research.

> These four references collectively define the competitive landscape — neural-network-based search, parallel search, and Monte Carlo search — against which the paper's contribution is implicitly positioned.

ORIGINAL (ENGLISH)

[8] C. Godsil, G. Royle, Algebraic Graph Theory, Vol. 207 of Graduate Texts in Mathematics, Springer, 2001. doi:10.1007/978-1-4613-0163-9.

EXPLANATION (ENGLISH)

This reference supports the **spectral graph theory** foundations used throughout the paper's graph-based analyses.

- *Godsil & Royle*: standard graduate textbook on algebraic/spectral graph methods

The capstone citation — like a bibliography card pointing to the textbook that taught the tools used in every equation before this.

Algebraic Graph Theory provides the eigenvalue and adjacency matrix machinery underlying all spectral arguments in this work.